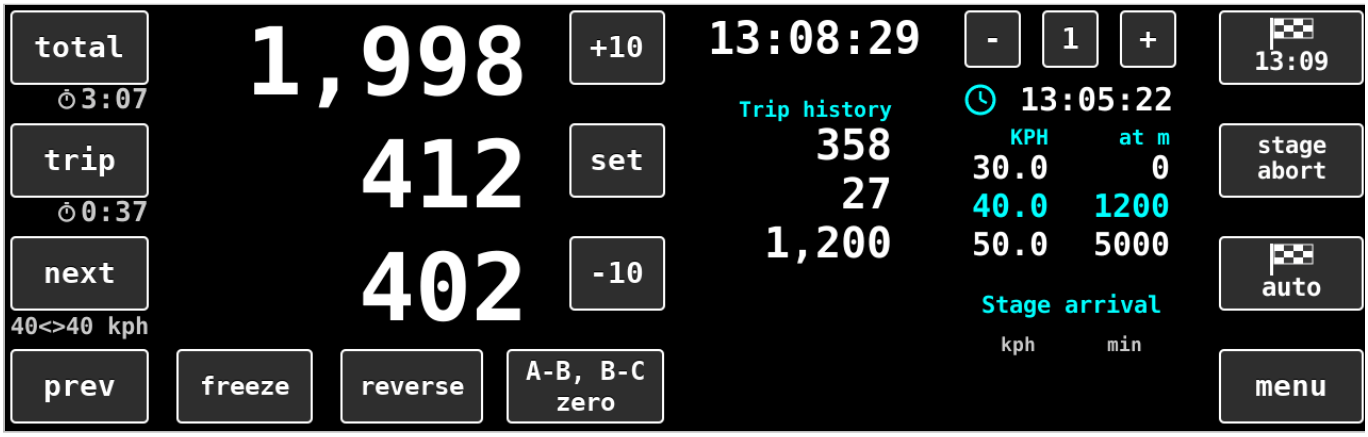
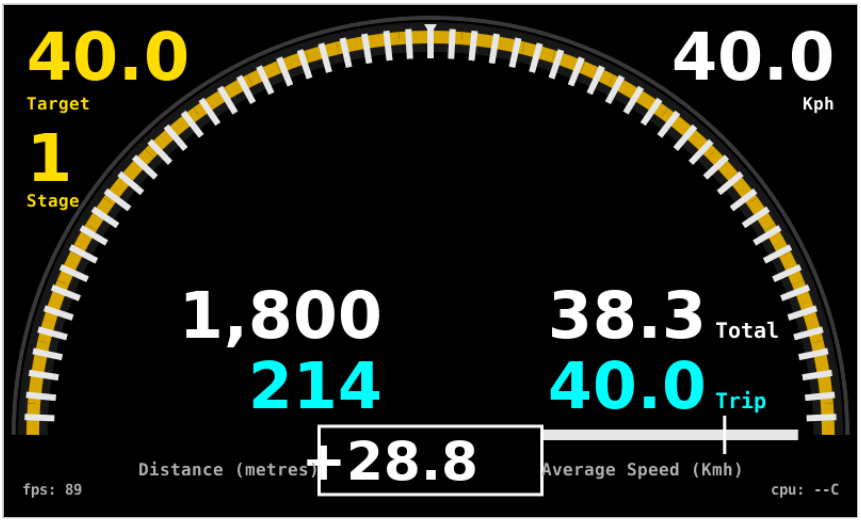
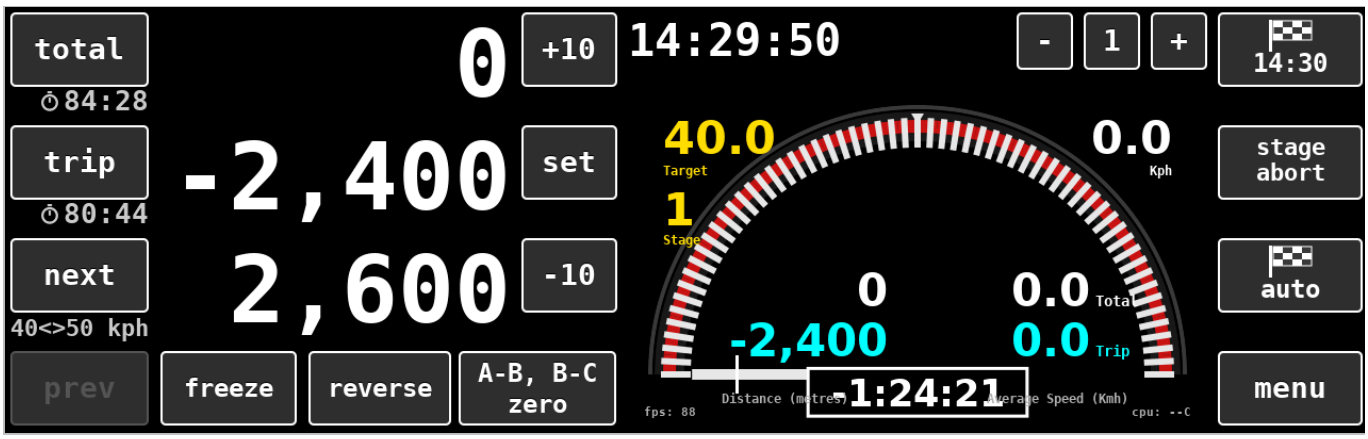
Rallybox User Manual



Co-pilot display, navigation page.



Driver display.



Single-display mode: the gauge sits on the co-pilot screen.

GETTING AROUND



The menu page.

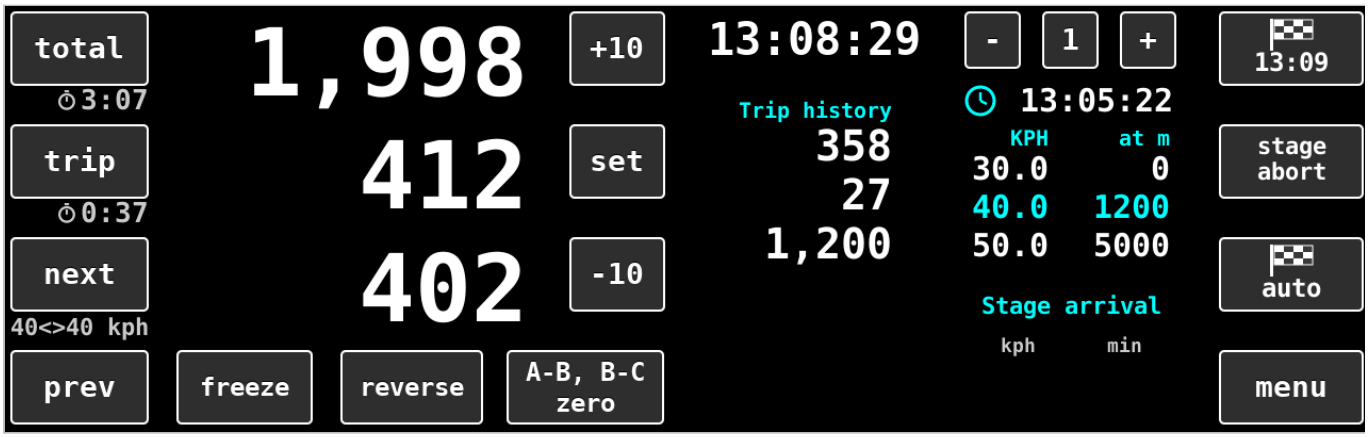
Every page has a `menu` key, always directly under the keypad (on the navigation page it is the bottom key of the right-hand column). Press it to reach the menu, then press the page you want. `navigation` returns to the main page.

- **navigation** – the main page, used while driving.
- **segments** – stage setup.
- **date / time** – the rally clock.
- **calibration** – pulses per kilometre.
- **settings** – beep, tone, units, logging and the phone link.
- **exit** – saves the settings and closes the program. It does not switch the Raspberry Pi off.

Numbers are typed on the on-screen keypad: tap a box, then use the keys. `C` clears the box and the arrow key deletes one character. The box has no Enter key, so an entry is saved when you tap another box or leave it alone for a second.

Speeds are shown in kph or mph, as chosen under settings (speeds in the stage table and the small `40<>50 kph` line follow the choice). The stage setup entries are always kph.

THE NAVIGATION PAGE



total

Shows the distance and elapsed time since the start. Pressing it resets the distance to zero.

- Outside a stage it resets the distance and the timer, but not Trip.
- Waiting for a timed autostart it resets Total, Trip and the timer at once, with no question; the stage zeroes everything again at its start time.
- Waiting for an early departure it resets Total and Trip at once, with no question, and leaves the timer alone: the minute starts it.
- During a stage you are asked to confirm (a missed start). Total and Trip go to zero from the moment you pressed, and the stage clock keeps running.

trip

An intermediate distance and time. Pressing it resets Trip, and the figure it showed is added to the **Trip history** column. Trip also restarts by itself at each speed change, whether the box makes it or `next` does, so it reads the distance driven in the present leg. `prev` puts Trip back to what it read before the change.

next and prev

The `next` figure is the distance still to run before the speed changes; below the key, `40<>50 kph` shows the present speed and the speed it changes to. `---` means there is no further change or no stage is running.

Press `next` at the speed change when the stage is not set to change by itself, or to move on early. What this section gives up is handed to the next, so the following change point stays where the roadbook puts it. If you press it after the next leg's whole length has gone (a missed change point), there is nothing left to take: the stage simply runs long and later change points move out. `next` works only during a stage, and does nothing after the last speed change. `prev` undoes the last speed change (whether the crew or the box made it); it is grey when there is nothing to undo.

+10, set and −10

Correct the measured distance. `+10` and `−10` move Total and Trip by ten metres a press. `set` opens a box to type the exact Total figure (Trip is not affected). Corrections during a stage re-work the average speeds and the timing error.

freeze and reverse

`freeze` holds the Total, Trip and next figures and the Trip history while it is lit, so they can be read or copied; the clock and the stage panel carry on. Freeze is shared: pressing it on the box or on a phone holds the box and every

phone together, and pressing it on either releases both. `reverse` makes Total, Trip and next count down while the car backs up; speed is unaffected. While either is lit the box repeats a low tone as a reminder.

A-B, B-C zero

Moves the driver's needle back to zero when the driver is deliberately aiming at an error (see driver zero).

The stage panel

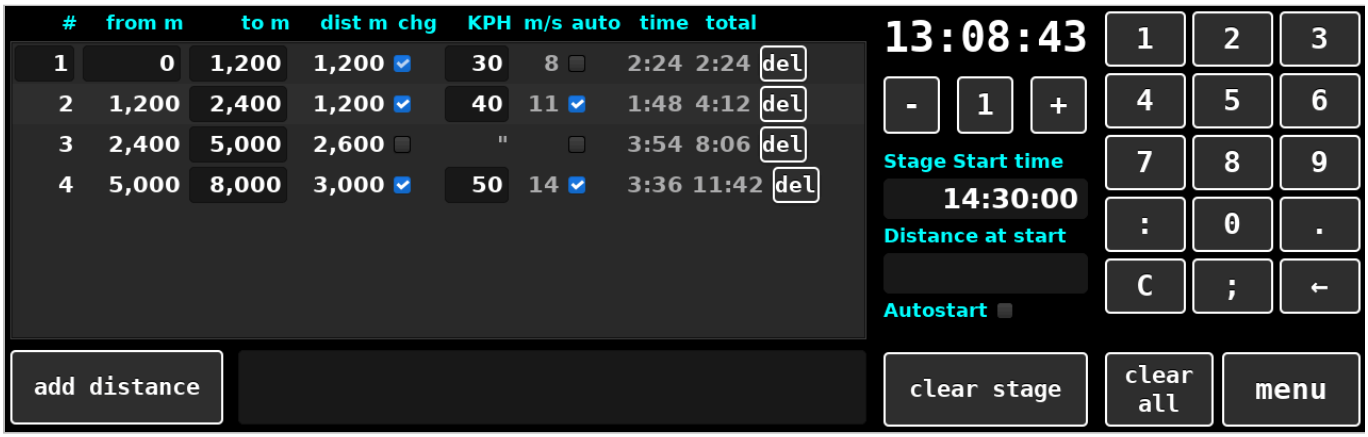
- **– 1 +** steps through the saved stages (1 to 5; empty ones are skipped). Tap the number to pick one directly. A new stage cannot be loaded while one is running.
- Beside it, a cyan clock and a time mean an autostart is armed for that time. A red cross means none is armed; a time with a red line through it has already gone by.
- The table lists the speed changes only, with the distance each begins at. The one in force is lit.
- **Stage arrival** gives the speed needed to reach the start a minute early and the minutes to go. It follows the loaded stage's start time and distance from the Segments page, and goes blank when under a minute remains.

Starting and ending a stage

The three flag keys at the right start the stage; the loaded stage is locked while it runs.

- **Flag with a time (for example 13:06)** – autostart on the next whole minute, shown on the key. Early departure is allowed: the distance zeroes and the stage loads when you press, so the driver display shows its speeds during the countdown; the timer zeroes at the minute. Anything rolled before the minute counts toward the stage distance and average speed. Pressing Total while waiting zeroes the distance and Trip again. If a stage is running you are asked first whether to abort it.
- **Flag with now** – starts the stage immediately, zeroing Total, Trip and the timer. Use it for a flying start. Once a stage is armed or running this key reads **stage abort** and, after a confirmation, ends the stage: the needle returns to zero, the tone stops, Total and Trip keep their readings and any autostart is cancelled. (A timed start still waiting does not change the key: nothing is live yet.)
- **Flag with auto** – arms an autostart at the loaded stage's start time (set on the Segments page). It turns white while armed; press it again to cancel. It is refused while a stage is running, or if the start time has passed. Early departure is not allowed with this start. (The Autostart tick on the Segments page can arm one during a running stage; that stage carries on until the autostart fires.)

SEGMENTS



The segments page.

Stages are kept in five memories, chosen with **– 1 +** under the clock. Everything typed is saved to the stage on show about a second after you stop typing. A stage cannot be edited while it is running.

Entering a stage

1. Tap the box beside `add distance` and type the distance at which each leg ends, as a roadbook total, separated by semicolons, for example `1200;2400;5000;8000`. Each must be larger than the one before.
2. Press `add distance`; the rows appear in the table.
3. Tick `chg` on each row where the speed changes and type the new speed in the KPH box. A row without a tick carries the speed on (shown as ").
4. Tick `auto` to have the box move to the next speed by itself at the end of that row; leave it clear to press `next`. The box is only available where there is a speed change.

The first row's # and `from m` can be typed over to match the roadbook; changing `from m` moves every row's `from m` and `to m` together without changing any leg's length. A `to m` figure can be corrected in place: the boundary moves and the next row takes up the difference, so later rows do not move. It must stay past that row's start and short of the next row's end; on the last row it simply lengthens or shortens the stage. The table also shows each leg's length, the speed in m/s, the time and the running total time. `del` removes a row and hands its length to the row above; deleting the first row moves `from m` up by its length instead.

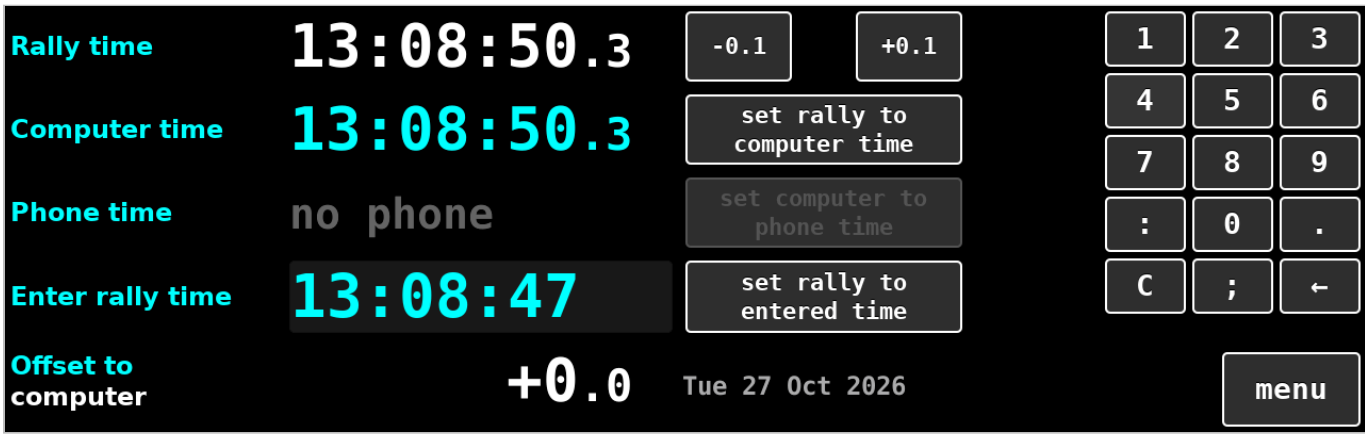
The stage column

- **Stage Start time** – the time the stage starts (hh:mm:ss).
- **Distance at start** – the roadbook distance at the start line; used for Stage arrival on the navigation page.
- **Autostart** – tick to arm an autostart at the start time today (a time already past is refused). Typing a time only stores it.
- `clear stage` empties this stage; `clear all` empties every stage. Both ask first.

Beep Assist

The `to m` figures double as the roadbook boxes for Beep Assist, so nothing more needs entering. Switch the beeps on under settings. Navigation gives a double beep a set number of metres before each box. Timing gives a single beep when the car should be passing it, and shows up distance errors from wheel spin, corner cutting or a skipping sensor. The beeps follow the Total reading, which is zero at the start of the stage, so a correction made with `+10`, `−10` or `set` moves every beep with it.

DATE / TIME



The date / time page.

The **Rally time** (white) is the clock every stage is timed against. The **Computer time** (cyan) is the Raspberry Pi's own clock. **Phone time** is the linked phone's clock, or `no phone`. **Offset to computer** is how far the rally clock is ahead (+) or behind (–) the computer; it is shown as `+2.3`, or `−11:49.2` once beyond a minute.

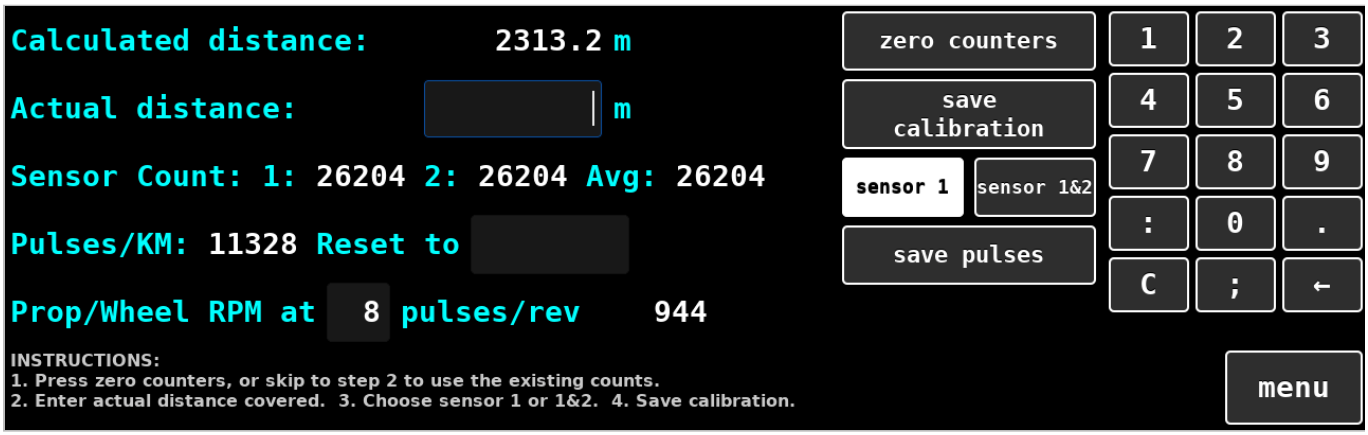
Setting the rally clock

- `set rally to computer time` makes the two the same.
- Type the time in the **Enter rally time** box as `hh:mm:ss` (or `hh:mm`), then press `set rally to entered time`. Today's date is used. Do not enter tenths.
- `set computer to phone time` sets the Pi's clock from the phone while keeping the rally clock where it is. It is grey until a phone is connected. It needs the Pi's usual permission to set its clock without a password; if it cannot, the box says `The computer clock could not be set` and nothing changes.

Trimming

`−0.1` and `+0.1` adjust the rally clock a tenth of a second at a time. Hold to repeat. Each press is saved at once.

CALIBRATION



The calibration page.

Calibrating over a set distance

1. Press `zero counters`, or skip this step to use the distance counted since the last Total reset.
2. Drive the measured distance (from 500 m to 100,000 m).
3. Type the **actual** distance driven in the Actual distance box, not the distance the box shows.
4. Press `sensor 1` or `sensor 1&2` to choose which counter is used; the choice is shown in white.
5. Press `save calibration`.

The page shows the calculated distance, each sensor's count and the pulses per kilometre now in use. If the actual distance is outside the range, or nothing has been counted, `save calibration` does nothing and gives no message, so check that the pulses per kilometre figure has changed.

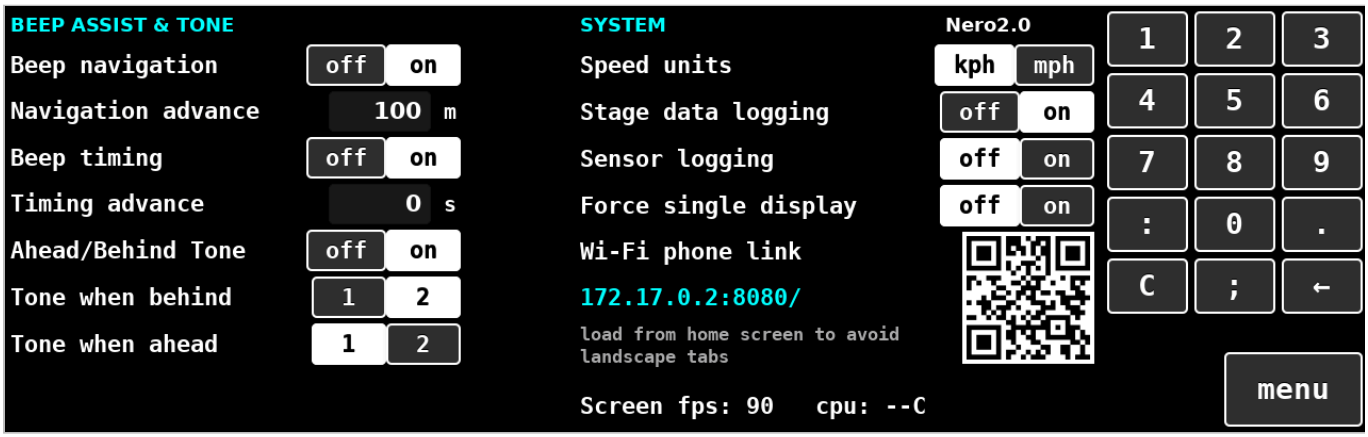
Setting the pulses directly

If the figure is already known, type it into the **Reset to** box and press `save pulses`. No driving is needed. Zero or less is ignored.

Prop/Wheel RPM

The last row shows propshaft or wheel revolutions per minute from sensor 1. Type the pulses the sensor gives per revolution into the small box for it to read correctly.

SETTINGS



The settings page.

Each switch shows its present state lit. Changes are saved at once.

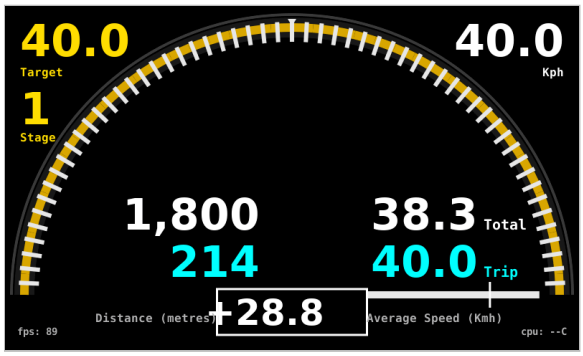
Beep Assist & Tone

- **Beep navigation** – off/on. **Navigation advance** is how many metres before a roadbook box the double beep sounds (100 is sensible).
- **Beep timing** – off/on. **Timing advance** is how many seconds before the box time the single beep sounds (0 is sensible).
- **Ahead/Behind Tone** – the driver's error tone, off/on.
- **Tone when behind** and **Tone when ahead** – each is type 1 or 2, and both tones stay silent for errors under 0.2 s and in the first 250 m. **Type 1** is a short buzzer tick. The bigger the error, the faster the ticking, in steps of 0.2 s: about one tick a second at 0.2 s, three a second at 1 s and about eight a second by 3 s. Behind is the slightly higher tick, ahead the lower. **Type 2** is one steady tone whose pitch rises with the error, in steps of 0.2 s up to 3 s: behind is the higher, harsher pitch, ahead the lower, smoother one.

System

- **Speed units** – kph or mph.
- **Stage data logging** – records a file for each stage, one row a second (distance, speed and ahead/behind error). It starts when the stage starts and stops when the stage ends or is aborted; switching it on during a stage starts a file at once. If the box cannot write the file it switches stage logging off by itself, without a message, and carries on.
- **Sensor logging** – records the raw counter readings continuously while it is on, many rows a second, so its files are much larger. Both kinds of log are deleted after 60 days, when the program starts, and can be downloaded from the phone (Set tab).
- **Force single display** – when on, the driver's gauge is placed on the co-pilot screen the next time the program starts, and the driver display is not used.
- **Wi-Fi phone link** – the address, and a QR code, to open the phone app. Part 2 describes it. The note beside it advises adding the page to the phone's home screen so it runs without the browser's bars.
- The foot of the column shows the screen update rate and the Pi's temperature.

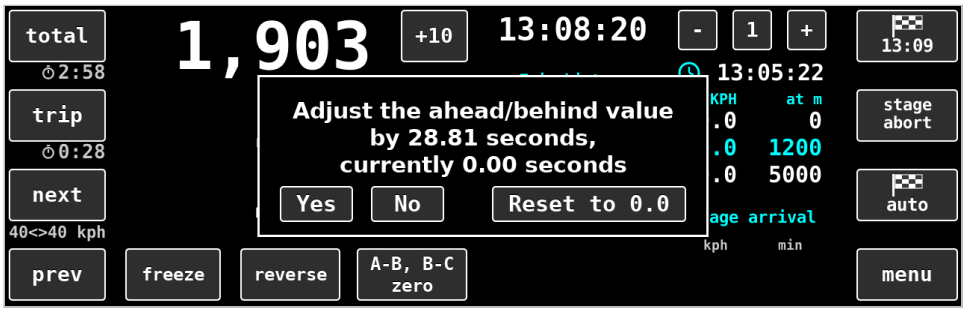
THE DRIVER DISPLAY AND A-B, B-C ZERO



The gauge shows the **Target** speed (yellow) and the stage number at top left, the present speed at top right, and the ahead/behind needle with the error in seconds in the box beneath it. Total and Trip distance, and their average speeds, are at the foot.

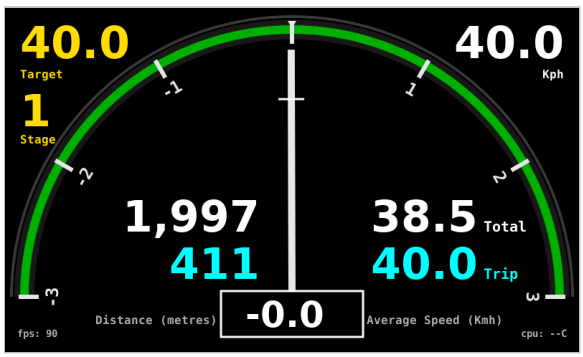
While an autostart is pending a T− `hh:mm:ss` box appears in the middle of the display, counting down to the start, with **Early Departure ENABLED** or **DISABLED** beneath it. A start on the next minute shows it all the way; a start set for later shows it only for the last five minutes. With early departure enabled the error and the average speeds are held at zero until the minute arrives, when the box disappears and the clock starts.

The `A−B, B−C zero` key rebases the needle when the driver is aiming at an error other than zero. The popup shows the error to be taken off and the adjustment already in force.



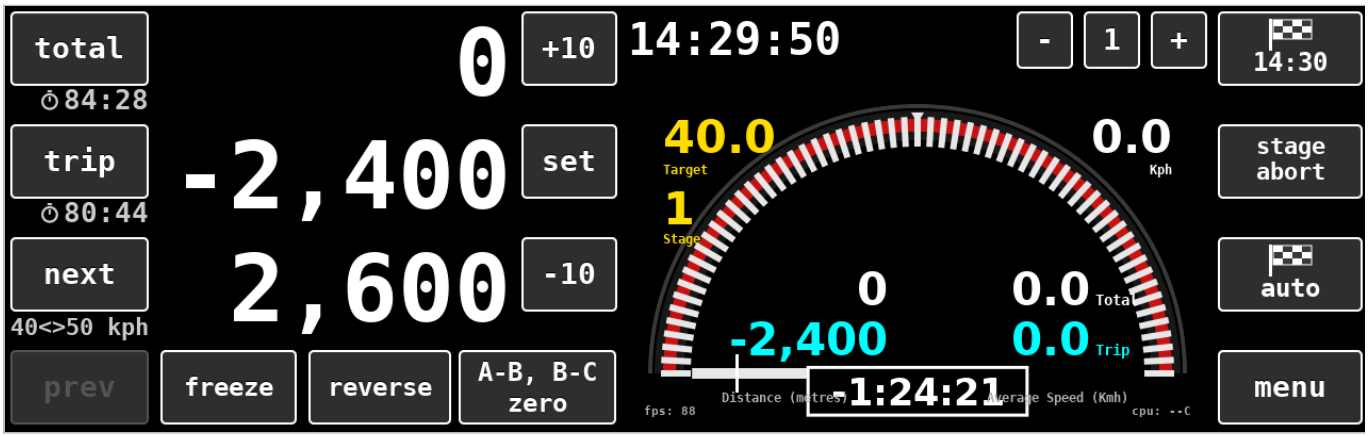
- `Yes` zeroes the needle at this instant.
- `Reset to 0.0` removes the adjustment so the true error shows.
- `No` changes nothing.

Only the driver's gauge changes; distances, schedule and clock are untouched. Starting a stage clears the adjustment.



After pressing Yes.

SINGLE-DISPLAY MODE



With **Force single display** on, the driver's gauge shares the co-pilot screen and there is only one screen. The keys work as described for the navigation page. The layout is: the Total / Trip / next block with `+10`, `set` and `–10` on the left, with `prev`, `freeze`, `reverse` and `A–B, B–C zero` along the foot; the rally clock and the **– 1 +** stage selector above the gauge in the middle; and the three flag keys and `menu` in a column on the right. There is no Trip history or stage table on this screen; segments, date / time and calibration are reached through `menu`, as on two displays. While a start is pending, the countdown box appears over the gauge.

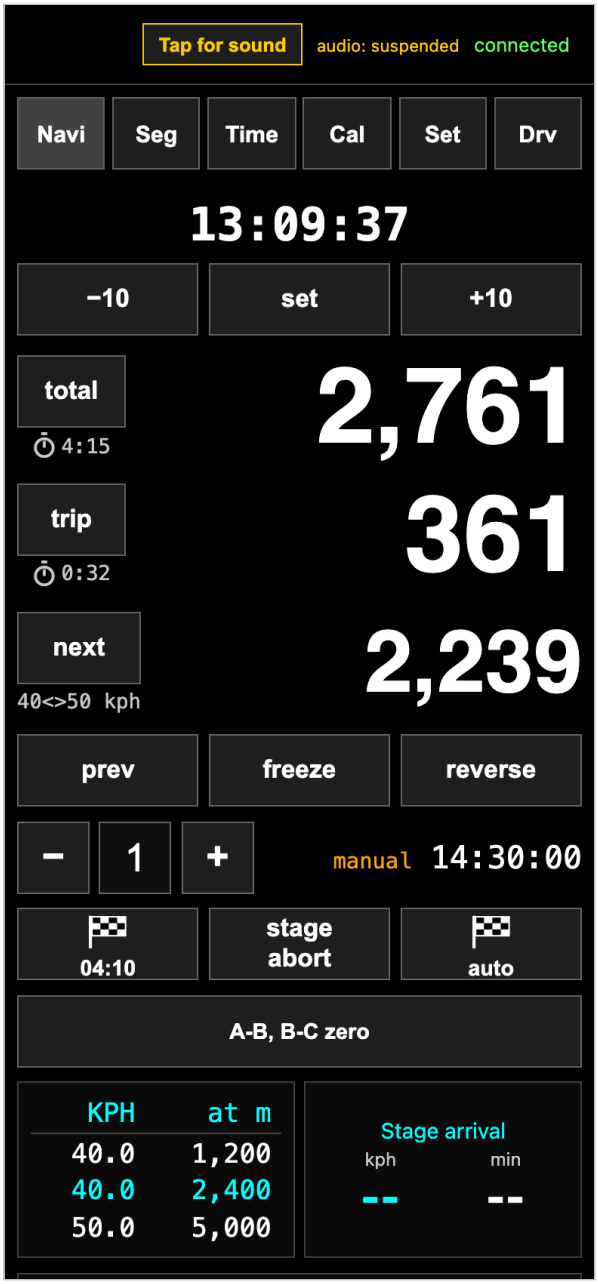
THE PHONE APP

Connecting: tether the rally box to the phone over Wi-Fi, then open the address shown on the settings page (or scan its QR code). The top of the page reads **connected** in green when linked. Adding the page to the phone's home screen runs it full screen.

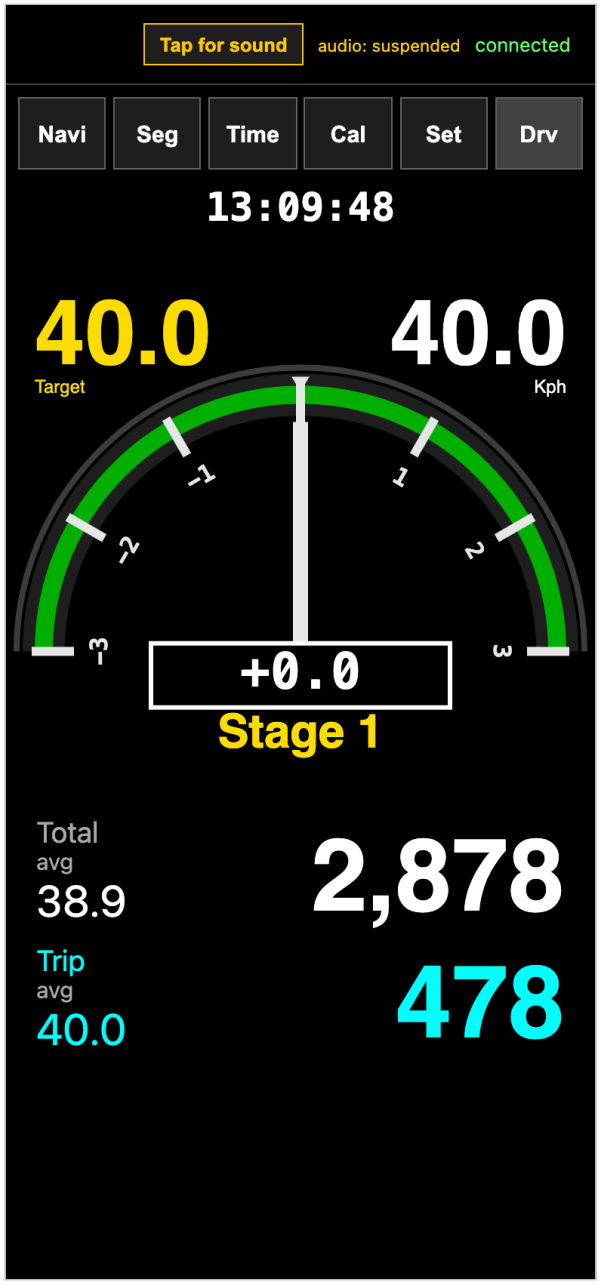
Sounds: the top key reads **Tap for sound** until the phone's audio is allowed; afterwards it toggles Sound is on / Sound is off.

Six tabs mirror the box: **Navi** (navigation, including total, trip, freeze, reverse, the three flag keys and A-B, B-C zero), **Seg** (segments), **Time** (date / time), **Cal** (calibration), **Set** (settings, plus `download logs`) and **Drv** (the driver gauge with Total and Trip). On Navi and Drv, press Total or Trip to reset them.

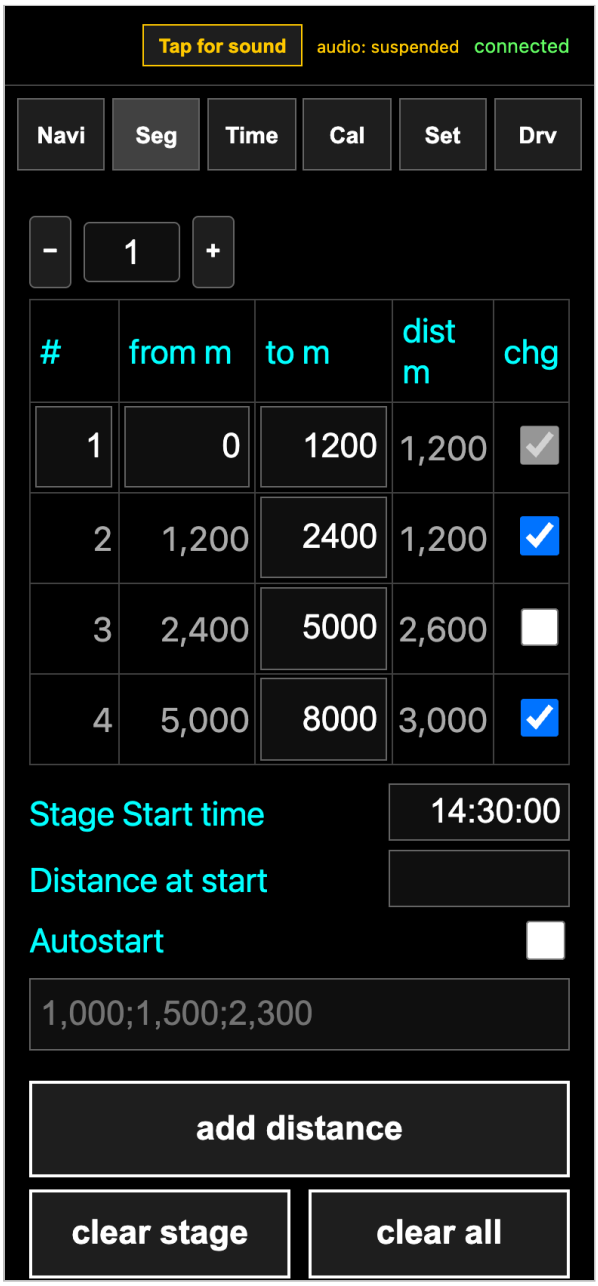
Navi



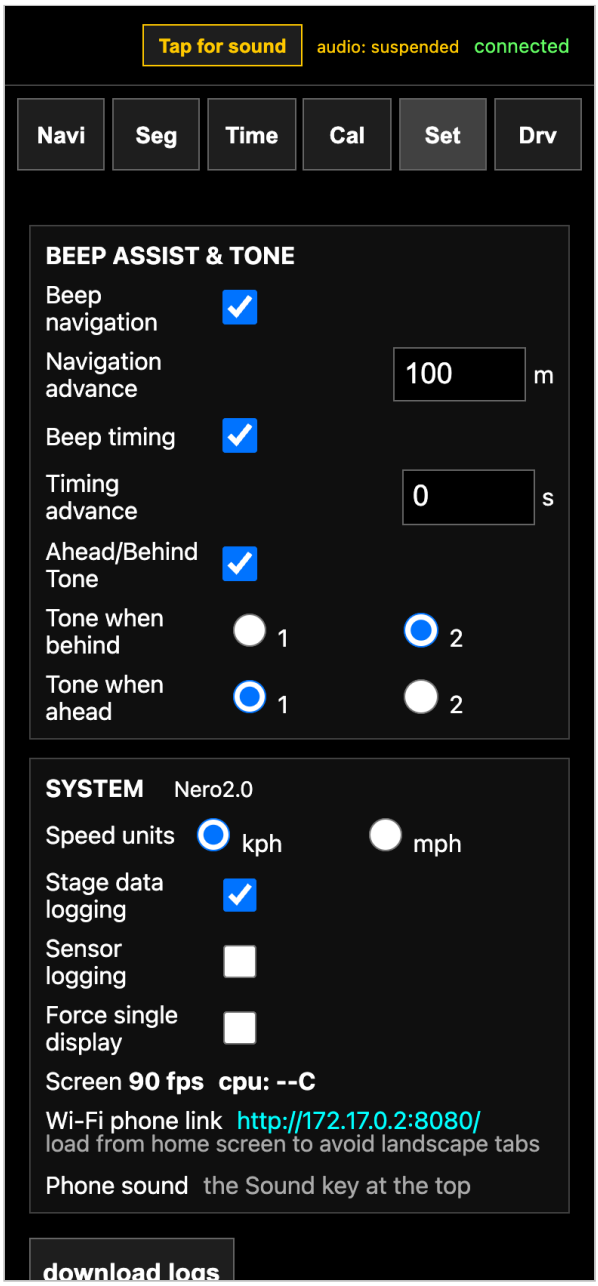
Drv



Seg


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
-
1
+
from m to m dist m chg
1 0 1200 1,200
2 1,200 2400 1,200
3 2,400 5000 2,600
4 5,000 8000 3,000
Stage Start time
14:30:00
Distance at start
Autostart
1,000;1,500;2,300
add distance
clear stage
clear all

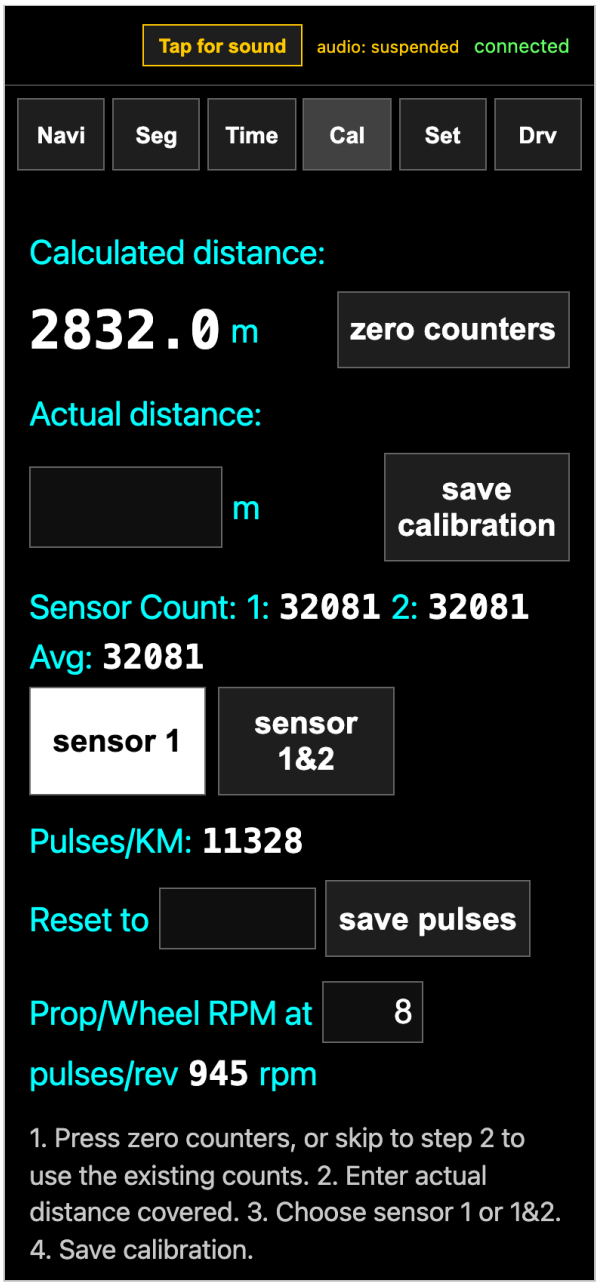
Set


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
BEEP ASSIST & TONE
Beep navigation
Navigation advance
100
m
Beep timing
Timing advance
0
s
Ahead/Behind Tone
Tone when behind
1
2
Tone when ahead
1
2
SYSTEM
Nero2.0
Speed units
kph
mph
Stage data logging
Sensor logging
Force single display
Screen 90 fps cpu: --C
Wi-Fi phone link http://172.17.0.2:8080/
load from home screen to avoid landscape tabs
Phone sound the Sound key at the top
download logs

Time


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
Rally time
13:09:42.2
−0.1
+0.1
Computer time
13:09:42.2
set rally to computer time
Phone time
13:09:42.2
set computer to phone time
Enter rally time
hh:mm:ss
set rally to entered time
Offset to computer
+0.0
Tue, 27 Oct 2026

Cal


Tap for sound
audio: suspended
connected
Navi
Seg
Time
Cal
Set
Drv
Calculated distance:
2832.0 m
zero counters
Actual distance:
m
save calibration
Sensor Count: 1: 32081 2: 32081
Avg: 32081
sensor 1
sensor 1&2
Pulses/KM: 11328
Reset to
save pulses
Prop/Wheel RPM at
8
pulses/rev 945 rpm
1. Press zero counters, or skip to step 2 to use the existing counts. 2. Enter actual distance covered. 3. Choose sensor 1 or 1&2. 4. Save calibration.